Problem 15.1.16

Is the series convergent or divergent? Give a reason.

$$\sum_{n=0}^{\infty} \frac{(20+30i)^n}{n!}$$

Solution

Ratio Test:

$$\begin{aligned}
\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n\to\infty} \left| \frac{(20+30i)^{n+1}}{(n+1)!} \cdot \frac{n!}{(20+30i)^n} \right| \\
&= \lim_{n\to\infty} \left| \frac{20+30i}{n+1} \right| \\
&= \lim_{n\to\infty} \frac{\sqrt{1300}}{n+1} \\
&= 0 < 1, \text{ therefore the series converges}
\end{aligned}$$

Problem 15.1.17

Is the series convergent or divergent? Give a reason.

$$\sum_{n=2}^{\infty} \frac{(-i)^n}{\ln n}$$

Solution

The numerator is bounded cyclically so only the denominator affects the series convergence.

$$\frac{1}{\ln n} > \frac{1}{n} \ \forall n \in \mathbb{Z} \geq 2$$

Therefore, the series diverges by direct comparison to the divergent harmonic series.

Problem 15.1.18

Is the series convergent or divergent? Give a reason.

$$\sum_{n=1}^{\infty} n^2 \left(\frac{i}{4}\right)^n$$

Solution

Ratio Test:

$$\begin{aligned}
\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| &= \lim_{n\to\infty}\left|\frac{(n+1)^2\left(\frac{i}{4}\right)^{n+1}}{n^2\left(\frac{i}{4}\right)^{n}}\right| \\
&= \lim_{n\to\infty}\left|\frac{(n+1)^2}{n^2}\cdot\frac{i}{4}\right| \\
&= \frac{1}{4}\cdot\lim_{n\to\infty}\frac{(n+1)^2}{n^2} \\
&= \frac{1}{4}\cdot 1 \\
&= \frac{1}{4} < 1, \text{ therefore the series converges}
\end{aligned}$$

Problem 15.1.19

Is the series convergent or divergent? Give a reason.

$$\sum_{n=0}^{\infty}\frac{i^n}{n^2-i}$$

Solution

$$|a_n| = \left|\frac{i^n}{n^2-i}\right| = \frac{1}{\sqrt{n^4+1}}$$

$$|a_n| < \frac{1}{n^2}\ \forall n \in \mathbb{Z} > 0$$

Therefore, the series converges by direct comparison to the convergent p-series $\sum_{n=0}^{\infty}\frac{1}{n^2}$.

Problem 15.1.20

Is the series convergent or divergent? Give a reason.

$$\sum_{n=0}^{\infty}\frac{n+i}{3n^2+2i}$$

Solution

The real part diverges due to the harmonic series and therefore the series diverges.

$$\sum_{n=0}^{\infty}\frac{n}{3n^2} = \frac{1}{3}\sum_{n=0}^{\infty}\frac{1}{n}$$

Problem 15.1.21

Is the series convergent or divergent? Give a reason.

$$\sum_{n=0}^{\infty}\frac{(\pi+\pi i)^{2n+1}}{(2n+1)!}$$

Solution

Ratio Test:

$$\begin{aligned}
\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| &= \lim_{n\to\infty}\left|\frac{(\pi+\pi i)^{2n+3}}{(2n+3)!}\cdot\frac{(2n+1)!}{(\pi+\pi i)^{2n+1}}\right| \\
&= \lim_{n\to\infty}\left|\frac{(\pi+\pi i)^2}{(2n+3)(2n+2)}\right| \\
&= 0 < 1, \text{ therefore the series converges}
\end{aligned}$$

Problem 15.1.22

Is the series convergent or divergent? Give a reason.

$$\sum_{n=1}^{\infty}\frac{1}{\sqrt{n}}$$

Solution

$$\frac{1}{\sqrt{n}} > \frac{1}{n}\ \forall n \in \mathbb{Z} > 1$$

Therefore, the series diverges by direct comparison with the harmonic series.

Problem 15.1.23

Is the series convergent or divergent? Give a reason.

$$\sum_{n=0}^{\infty}\frac{(-1)^n(1+i)^{2n}}{(2n)!}$$

Solution

Ratio Test:

$$\begin{aligned}
\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| &= \lim_{n\to\infty}\left|\frac{(-1)^{n+1}(1+i)^{2(n+1)}}{(2(n+1))!}\cdot\frac{(2n)!}{(-1)^n(1+i)^{2n}}\right| \\
&= \lim_{n\to\infty}\left|\frac{-(1+i)^2}{(2n+2)(2n+1)}\right| \\
&= 0 < 1, \text{ therefore the series converges}
\end{aligned}$$

Problem 15.1.24

Is the series convergent or divergent? Give a reason.

$$\sum_{n=1}^{\infty}\frac{(3i)^n n!}{n^n}$$

Solution

Ratio Test:

$$\begin{aligned}
\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| &= \lim_{n\to\infty}\left|\frac{(3i)^{n+1}(n+1)!}{(n+1)^{n+1}}\cdot\frac{n^n}{(3i)^n n!}\right| \\
&= \lim_{n\to\infty}\left|\frac{(3i)^{n+1}}{(3i)^n}\cdot\frac{(n+1)!}{n!}\cdot\frac{n^n}{(n+1)^{n+1}}\right| \\
&= \lim_{n\to\infty}\left|3i\cdot(n+1)\cdot\frac{n^n}{(n+1)^n(n+1)}\right| \\
&= 3\lim_{n\to\infty}\left|\left(\frac{n}{n+1}\right)^n\right| \\
&= 3\lim_{n\to\infty}\left|\frac{1}{\left(\frac{n+1}{n}\right)^n}\right| \\
&= 3\lim_{n\to\infty}\left|\frac{1}{\left(1+\frac{1}{n}\right)^n}\right| \\
&= 3\cdot\frac{1}{e} \\
&= \frac{3}{e} > 1, \text{ therefore the series diverges}
\end{aligned}$$

Problem 15.2.7

Find the center and the radius of convergence.

$$\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n)!}\left(z-\frac{1}{2}\right)^{2n}$$

Solution

$$\begin{aligned}
R &= \lim_{n\to\infty}\left|\frac{a_n}{a_{n+1}}\right| \\
&= \lim_{n\to\infty}\left|\frac{(-1)^n}{(2n)!}\cdot\frac{(2n+2)!}{(-1)^{n+1}}\right| \\
&= \lim_{n\to\infty}\left|(2n+2)(2n+1)\right| \\
&= \infty
\end{aligned}$$

The center of convergence is at $z_0 = \frac{1}{2}$ with radius $R = \infty$.

Problem 15.2.9

Find the center and the radius of convergence.

$$\sum_{n=0}^{\infty}\frac{n(n-1)}{3^n}(z-i)^{2n}$$

Solution

Rearranging the power series to the classic form by substituting $n = 2n$,

$$\sum_{n=0}^{\infty} \frac{\frac{n}{2}(\frac{n}{2}-1)}{3^{\frac{n}{2}}}(z-i)^n = \frac{1}{4}\sum_{n=0}^{\infty} \frac{n(n-2)}{3^{\frac{n}{2}}}(z-i)^n$$

$$\begin{aligned} R &= \lim_{n\to\infty}\left|\frac{a_n}{a_{n+1}}\right| \\ &= \lim_{n\to\infty}\left|\frac{n(n-2)}{3^{\frac{n}{2}}}\cdot\frac{3^{\frac{n+1}{2}}}{(n+1)(n-1)}\right| \\ &= \sqrt{3}\lim_{n\to\infty}\left|\frac{n(n-2)}{(n+1)(n-1)}\right| \\ &= \sqrt{3} \end{aligned}$$

The center of convergence is at $z_0 = i$ with radius $R = \sqrt{3}$.

Problem 15.2.13

Find the center and the radius of convergence.

$$\sum_{n=0}^{\infty} 16^n (z+i)^{4n}$$

Solution

Rearranging the power series to the classic form by substituting $n = 4n$,

$$\sum_{n=0}^{\infty} 16^{\frac{n}{4}}(z+i)^n$$

$$\begin{aligned} R &= \lim_{n\to\infty}\left|\frac{a_n}{a_{n+1}}\right| \\ &= \lim_{n\to\infty}\left|\frac{16^{\frac{n}{4}}}{16^{\frac{n+1}{4}}}\right| \\ &= \lim_{n\to\infty}\left|16^{-\frac{1}{4}}\right| \\ &= \frac{1}{2} \end{aligned}$$

The center of convergence is at $z_0 = -i$ with radius $R = \frac{1}{2}$.

Problem 15.2.15

Find the center and the radius of convergence.

$$\sum_{n=0}^{\infty} \frac{(2n)!}{4^n (n!)^2}(z-2i)^n$$

Solution

$$\begin{aligned}
R &= \lim_{n\to\infty}\left|\frac{a_n}{a_{n+1}}\right| \\
&= \lim_{n\to\infty}\left|\frac{(2n)!}{4^n(n!)^2}\cdot\frac{4^{n+1}((n+1)!)^2}{(2n+2)!}\right| \\
&= \lim_{n\to\infty}\left|4\cdot\frac{(2n)!}{(2n+2)!}\cdot\frac{((n+1)!)^2}{(n!)^2}\right| \\
&= \lim_{n\to\infty}\left|4\cdot\frac{1}{(2n+2)(2n+1)}\cdot(n+1)^2\right| \\
&= \lim_{n\to\infty}\left|\frac{2n+2}{2n+1}\right| \\
&= 1
\end{aligned}$$

The center of convergence is at $z_0 = 2i$ with radius $R = 1$.

Problem 15.2.17

Find the center and the radius of convergence.

$$\sum_{n=1}^{\infty}\frac{2^n}{n(n+1)}z^{2n+1}$$

Solution

$$\begin{aligned}
L &= \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| \\
&= \lim_{n\to\infty}\left|\frac{2^{n+1}}{(n+1)(n+2)}z^{2n+3}\cdot\frac{n(n+1)}{2^n}z^{-(2n+1)}\right| \\
&= \lim_{n\to\infty}\left|2z^2\cdot\frac{n}{n+1}\right| \\
&= 2\left|z\right|^2
\end{aligned}$$

$$\begin{aligned}
L &< 1 \\
2\left|z\right|^2 &< 1 \\
\left|z\right|^2 &< \frac{1}{2} \\
\left|z\right| &< \frac{1}{\sqrt{2}}
\end{aligned}$$

The center of convergence is at $z_0 = 0$ with radius $R = \frac{1}{\sqrt{2}}$.

Problem 15.3.5

Find the radius of convergence in two ways:

(a) Directly by the Cauchy–Hadamard formula in Sec. 15.2
(b) From a series of simpler terms by using Theorem 3 or Theorem 4

$$\sum_{n=2}^{\infty} \frac{n(n-1)}{2^n}(z-2i)^n$$

Solution

Cauchy-Hadamard

$$\begin{aligned}
R &= \lim_{n\to\infty} \left| \frac{a_n}{a_{n+1}} \right| \\
&= \lim_{n\to\infty} \left| \frac{n(n-1)}{2^n} \cdot \frac{2^{n+1}}{(n+1)n} \right| \\
&= 2 \cdot \lim_{n\to\infty} \left| \frac{n-1}{n+1} \right| \\
&= 2
\end{aligned}$$

Theorem 3: Differentiation

$$f(z) = \sum_{n=0}^{\infty} \left(\frac{z-2i}{2} \right)^n = \sum_{n=0}^{\infty} \frac{1}{2^n}(z-2i)^n$$

The geometric series $f(z)$ converges if

$$\frac{z-2i}{2} < 1 \implies R = 2$$

Differentiating,

$$\begin{aligned}
f'(z) &= \sum_{n=1}^{\infty} \frac{n}{2^n}(z-2i)^{n-1} \\
f''(z) &= \sum_{n=2}^{\infty} \frac{n(n-1)}{2^n}(z-2i)^{n-2}
\end{aligned}$$

The original series is equivalent to $f''(z)(z-2i)^2$ and therefore also converges with $R = 2$

Problem 15.3.7

Find the radius of convergence in two ways:

(a) Directly by the Cauchy–Hadamard formula in Sec. 15.2
(b) From a series of simpler terms by using Theorem 3 or Theorem 4

$$\sum_{n=1}^{\infty} \frac{n}{3^n}(z+2i)^{2n}$$

Solution

Cauchy-Hadamard

$$\begin{aligned}
L &= \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| \\
&= \lim_{n\to\infty}\left|\frac{3^{n+1}}{n+1}(z+2i)^{2n+2}\cdot\frac{n}{3^n}(z+2i)^{-2n}\right| \\
&= \frac{1}{3}\cdot\lim_{n\to\infty}\left|(z+2i)^2\cdot\frac{n+1}{n}\right| \\
&= \frac{1}{3}\left|z+2i\right|^2
\end{aligned}$$

$$\begin{aligned}
\frac{1}{3}\left|z+2i\right|^2 &< 1 \\
\left|z+2i\right|^2 &< 3 \\
\left|z+2i\right| &< \sqrt{3}
\end{aligned}$$

Therefore $R=\sqrt{3}$

Theorem 3: Differentiation

$$f(z)=\sum_{n=0}^{\infty}\left(\frac{(z+2i)^2}{3}\right)^n=\sum_{n=0}^{\infty}\frac{1}{3^n}(z+2i)^{2n}$$

The geometric series $f(z)$ converges if

$$\frac{(z+2i)^2}{3}<1 \implies R=\sqrt{3}$$

Differentiating,

$$f'(z)=\sum_{n=1}^{\infty}\frac{2n}{3^n}(z+2i)^{2n-1}$$

The original series is equivalent to $\frac{1}{2}(z+2i)f'(z)$ and therefore also converges with $R=\sqrt{3}$

Problem 15.3.9

Find the radius of convergence in two ways:

(a) Directly by the Cauchy–Hadamard formula in Sec. 15.2
(b) From a series of simpler terms by using Theorem 3 or Theorem 4

$$\sum_{n=1}^{\infty}\frac{(-2)^n}{n(n+1)(n+2)}z^{2n}$$

Solution

Cauchy-Hadamard

$$\begin{aligned} L &= \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| \\ &= \lim_{n\to\infty} \left| \frac{(-2)^{n+1}}{(n+1)(n+2)(n+3)} z^{2n+2} \cdot \frac{n(n+1)(n+2)}{(-2)^n} z^{-2n} \right| \\ &= 2\left|z\right|^2 \cdot \lim_{n\to\infty} \left| \frac{n(n+1)(n+2)}{(n+1)(n+2)(n+3)} \right| \\ &= 2\left|z\right|^2 \end{aligned}$$

$$\begin{aligned} L &< 1 \\ 2\left|z\right|^2 &< 1 \\ \left|z\right|^2 &< \frac{1}{2} \\ \left|z\right| &< \frac{1}{\sqrt{2}} \end{aligned}$$

Therefore $R = \frac{1}{\sqrt{2}}$

Theorem 4: Integration

$$f(z) = \sum_{n=0}^{\infty} \left(-2z^2\right)^n$$

The geometric series $f(z)$ converges if

$$\left|-2z^2\right| < 1 \implies R = \frac{1}{\sqrt{2}}$$

Integrating 3 times introduces the factors $\frac{1}{n+1}$, $\frac{1}{n+2}$, and $\frac{1}{n+3}$ which is similar to the original series and it therefore also converges with $R = \frac{1}{\sqrt{2}}$

Problem 15.3.11

Find the radius of convergence in two ways:

(a) Directly by the Cauchy–Hadamard formula in Sec. 15.2
(b) From a series of simpler terms by using Theorem 3 or Theorem 4

$$\sum_{n=1}^{\infty} \frac{3^n n(n+1)}{7^n} (z+2)^{2n}$$

Solution

Cauchy-Hadamard

$$\begin{aligned} L &= \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| \\ &= \lim_{n\to\infty}\left|\frac{3^{n+1}(n+1)(n+2)}{7^{n+1}}(z+2)^{2n+2}\cdot\frac{7^n}{3^n n(n+1)}(z+2)^{-2n}\right| \\ &= \frac{3}{7}|z+2|^2\cdot\lim_{n\to\infty}\left|\frac{(n+1)(n+2)}{n(n+1)}\right| \\ &= \frac{3}{7}|z+2|^2 \end{aligned}$$

$$\begin{aligned} L &< 1 \\ \frac{3}{7}|z+2|^2 &< 1 \\ |z+2|^2 &< \frac{7}{3} \\ |z+2| &< \sqrt{\frac{7}{3}} \end{aligned}$$

Therefore $R = \sqrt{\frac{7}{3}}$

Theorem 3: Differentiation

$$f(z) = \sum_{n=0}^{\infty}\left(\frac{3}{7}(z+2)^2\right)^n$$

The geometric series $f(z)$ converges if

$$\left|\frac{3}{7}(z+2)^2\right| < 1 \implies R = \sqrt{\frac{7}{3}}$$

Taking the derivative 2 times introduces the factors n and $n+1$ which is similar to the original series and it therefore also converges with $R = \sqrt{\frac{7}{3}}$

Problem 15.3.13

Find the radius of convergence in two ways:

(a) Directly by the Cauchy–Hadamard formula in Sec. 15.2
(b) From a series of simpler terms by using Theorem 3 or Theorem 4

$$\sum_{n=0}^{\infty}\left[\binom{n+k}{k}\right]^{-1} z^{n+k} = \sum_{n=0}^{\infty}\frac{k!n!}{(n+k)!}z^{n+k}$$

Solution

Cauchy-Hadamard

$$\begin{aligned}
L &= \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| \\
&= \lim_{n\to\infty}\left|\frac{k!(n+1)!}{(n+1+k)!}z^{n+1+k}\cdot\frac{(n+k)!}{k!n!}z^{-n-k}\right| \\
&= |z|\cdot\lim_{n\to\infty}\left|\frac{(n+1)!}{n!}\cdot\frac{(n+k)!}{(n+1+k)!}\right| \\
&= |z|\cdot\lim_{n\to\infty}\left|\frac{n+1}{n+k}\right| \\
&= |z|
\end{aligned}$$

$$L<1 \implies R=1$$

Theorem 3: Differentiation

$$f(z)=\sum_{n=0}^{\infty} z^n$$

The geometric series $f(z)$ converges if

$$|z|<1 \implies R=1$$

Integrating k times

$$\int\int\cdots\int f(z) \propto \frac{z^{n+k}}{(n+1)(n+2)\cdots(n+k)}$$

This gives a series that is proportional to the original and therefore also converges with $R=1$